

# Project 3: Active Learning for Learning-to-Defer

Human-Centric Artificial Intelligence | TU Hamburg | SoSe 2026

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## 1. Dataset and experimental setup

The project uses AG News, with four classes: World, Sports, Business and Sci/Tech. The supplied data contain 120,000 training articles and 7,600 test articles. We join the title and description into one text field and map the original labels 1-4 to 0-3. The test set is kept for evaluation only. The experiments first build a text classifier, then add a simulated expert, compare two learning-to-defer rules, and finally study how expert competence can be learned with a limited query budget.

## 2. Baseline classifier (Task 1)

We use TF-IDF features followed by logistic regression. The vectorizer uses unigrams and bigrams, removes English stop words and keeps at most 50,000 features. Logistic regression is trained with  $C = 1.0$  and a maximum of 1,000 iterations on the complete training set. This model was chosen because it is a strong and simple baseline for text classification and, importantly for the later tasks, it provides class probabilities.

| Class    | Precision | Recall | F1     |
|----------|-----------|--------|--------|
| World    | 0.9346    | 0.9100 | 0.9221 |
| Sports   | 0.9570    | 0.9837 | 0.9702 |
| Business | 0.8961    | 0.8900 | 0.8931 |
| Sci/Tech | 0.8990    | 0.9037 | 0.9013 |

Table 1. Baseline performance on the AG News test set.

The classifier predicts 7,006 of the 7,600 test articles correctly, giving a test accuracy of 92.18%. Performance is strongest on Sports and slightly lower on Business and Sci/Tech. This classifier is used unchanged in the remaining experiments.

## 3. Simulated expert (Task 2)

The simulated expert has a clear competence profile: it answers World and Sports correctly with probability 0.95, and Business and Sci/Tech with probability 0.55. When it is wrong, one of the other three labels is chosen uniformly. The random generator is seeded by article index, so the same article receives the same expert answer whenever it is queried. This makes comparisons between strategies reproducible.

| Class    | Designed competence | Observed test accuracy |
|----------|---------------------|------------------------|
| World    | 0.95                | 0.9537                 |
| Sports   | 0.95                | 0.9511                 |
| Business | 0.55                | 0.5589                 |
| Sci/Tech | 0.55                | 0.5626                 |

Table 2. Simulated expert strengths and weaknesses.

The expert reaches 75.66% overall test accuracy. It is therefore weaker than the classifier overall, but it is not useless: on some examples where the classifier fails, the expert is correct. The learning-to-defer task is to identify those cases without sending every article to the expert.

#### 4. Learning to defer (Task 3)

We compare two deferral strategies. The first is a confidence rule: defer when the classifier maximum probability is below a threshold. The threshold is tuned on a fixed sample of 30,000 training articles using the simulated expert labels; the selected value is 0.51.

The second strategy learns a separate logistic-regression policy. Its target is 1 when the classifier is wrong and the expert is correct. The inputs are the four classifier probabilities, maximum confidence, prediction entropy, and four indicators for the predicted class. The 30,000-example sample is split 70/30 for policy fitting and threshold tuning. The chosen decision threshold is 0.17.

| Method               | Accuracy | Defer rate | Helpful | Harmful | Helpful / deferred |
|----------------------|----------|------------|---------|---------|--------------------|
| Classifier only      | 92.18%   | 0.00%      | -       | -       | -                  |
| Expert only          | 75.66%   | 100.00%    | -       | -       | -                  |
| Confidence threshold | 92.54%   | 4.33%      | 89      | 62      | 27.05%             |
| Learned policy       | 92.58%   | 5.91%      | 120     | 90      | 26.73%             |

Table 3. Learning-to-defer results on the common test set.

A helpful deferral means the classifier is wrong and the expert is correct; a harmful deferral means the classifier is correct and the expert is wrong. The confidence rule makes 89 helpful and 62 harmful deferrals, a net gain of 27 correct predictions. The learned policy makes 120 helpful and 90 harmful deferrals, a net gain of 30. This corresponds to improvements of 0.355 and 0.395 percentage points over the classifier-only baseline. The learned policy is slightly better, but the difference between the two deferral strategies is small.

The learned policy also defers more often on articles predicted as Business and Sci/Tech than on World and Sports. This is reasonable because the classifier is less certain in these areas, although the simulated expert itself is also weaker on these two classes. The policy therefore has to balance classifier uncertainty against the expected value of asking the expert.

| Deferral outcome     | Confidence threshold | Learned policy |
|----------------------|----------------------|----------------|
| Helpful              | 89                   | 120            |
| Harmful              | 62                   | 90             |
| Both correct         | 121                  | 175            |
| Both wrong           | 57                   | 64             |
| Missed opportunities | 321                  | 290            |

Table 4. Breakdown of the deferral decisions on the test set.

| Classifier predicted class | Learned-policy defer rate |
|----------------------------|---------------------------|
| World                      | 2.86%                     |
| Sports                     | 3.84%                     |
| Business                   | 8.00%                     |
| Sci/Tech                   | 8.90%                     |

Table 5. Where the learned policy sends examples to the expert.

## 5. Active learning for expert competence discovery (Task 4)

For this part we assume that no expert labels are initially available. We sample a pool of 30,000 training articles and allow a total of 3,000 expert queries, added in batches of 300. We compare uncertainty sampling with uniform random sampling. Uncertainty sampling always selects the remaining articles with the lowest classifier maximum probability, since these are natural candidates for future deferral.

Because the simulated expert was defined by class-dependent competence, we estimate its accuracy separately for the four true topics from the queried examples. The evaluation metric is the mean absolute error (MAE) between the estimated four-class competence profile and the known simulated profile. Results are averaged over five different pool samples.

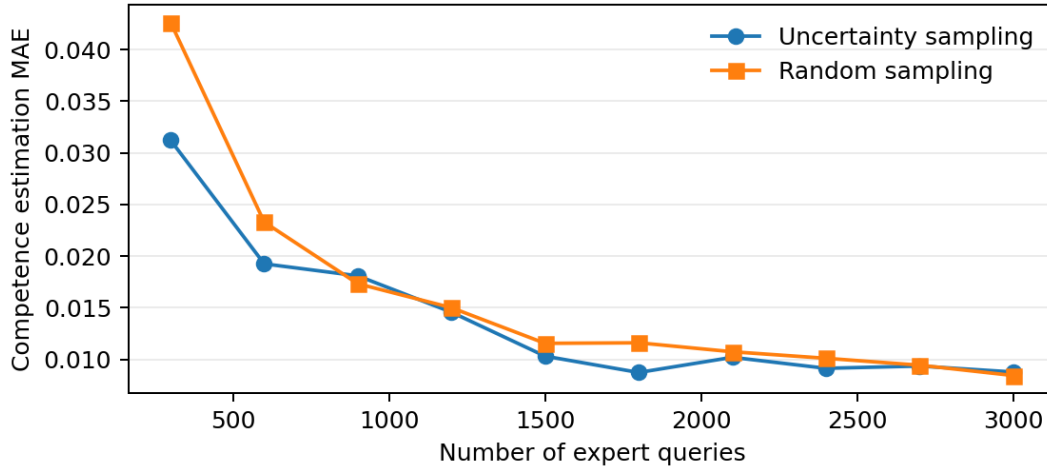


Figure 1. Competence-profile estimation error. Lower is better.

At 300 queries, the MAE is 0.0312 for uncertainty sampling and 0.0426 for random sampling, about 27% lower for the active strategy. Uncertainty sampling is also better at 600 queries (0.0192 versus 0.0233). The advantage does not remain at every later checkpoint: random sampling is slightly better at 900 queries and again at 3,000 queries (0.00843 versus 0.00877). The result therefore supports uncertainty sampling mainly as an early-budget strategy; with enough labels, both methods estimate this simple competence profile well.

## 6. Interface and limitations

The Django page presents the baseline, expert, deferral results and active-learning curve. As an optional extension, a participant can act as the expert: the interface shows articles selected by uncertainty sampling, hides the AG News label until an answer is submitted, and reports the participant's overall and per-topic accuracy for the current session.

There are two main limitations. First, Task 4 evaluates how well the expert competence profile is recovered; it does not retrain and evaluate a complete deferral policy after every query batch. Second, the probabilities used to train the Task 3 deferral policy come from examples that were also used to train the baseline classifier. Out-of-fold classifier probabilities would give a cleaner training signal for a future version. Within these limits, the experiments show that selective deferral can improve the strong classifier slightly, and that uncertainty sampling learns the expert profile faster at small query budgets.